

What chemistry faculty believe about doctoral education

A phenomenographic and thematic characterization

Zoe Rahimi Pirkoohi Jordan Harshman

University of Iowa



Doctoral education produces the chemists who staff the discipline.

~2,900

chemistry PhDs
awarded each year

~196

U.S. doctoral
programs

Decades

the same structure,
largely unchanged

The same record documents its strains — competing responsibilities, no standard assessment of outcomes, uneven implementation, and inconsistent mentorship.

Collini et al., 2025 · Donkor & Harshman, 2023

THE GAP

We know what faculty believe about *teaching a class*.

We don't know what they believe about *advising a PhD*.

CLASSROOM TEACHING

Faculty beliefs about teaching and learning are well characterized — and predict what happens in the classroom.

Gibbons · Lund & Stains · Popova · Mack & Towns

RESEARCH ADVISING

The actual work of the doctorate is a **different practice** — and beliefs are tied to the practice they concern. Never characterized.

Kagan, 1992

Donkor et al. (2024) set faculty beliefs deliberately aside — a proper investigation *“warranted its own study.”* This is that study.

RESEARCH QUESTION

What do chemistry faculty believe
about doctoral education?

What the doctorate is for



How students learn

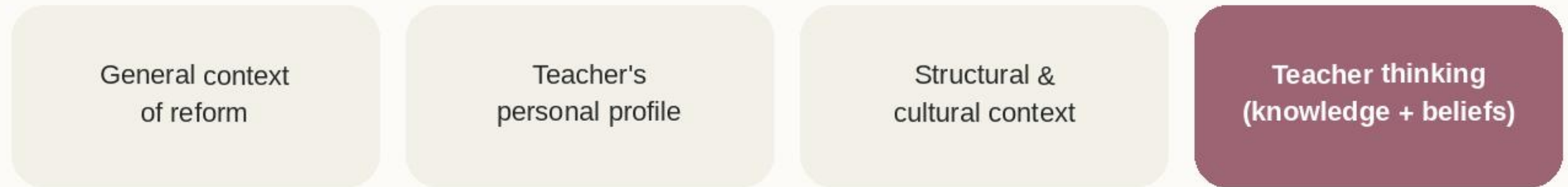


Who develops which skill



The structural conditions

Reforms keep changing the structures of schooling without changing what happens inside them.



↓ all four shape whether a reform reaches **practice** — the ultimate determinant of its success

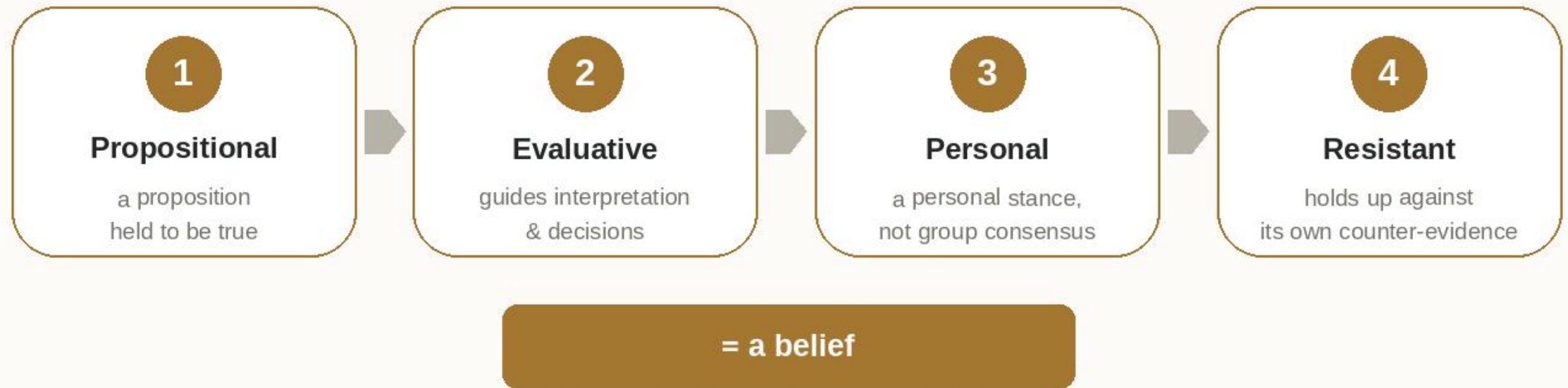
Beliefs are self-perpetuating — they can **absorb a reform without yielding to it.**

WHAT COUNTS AS A BELIEF

A *proposition a person holds true* and uses to interpret experience and guide action.

Rokeach, 1968 · Green, 1971 · Nespor, 1987 · Pajares, 1992

A unit of talk counts as a belief only if it passes all four tests:



METHODS

An interpretive design built to capture the *range* of faculty belief

- **20 faculty**

across U.S. doctoral programs, research-active, currently advising

- **Interviews + card sort**

~60-min semi-structured (modified Teacher Beliefs Interview); the sort as a second elicitation

- **Phenomenography + thematic analysis**

the range of ways faculty understand, plus keyness — not frequency — as the reporting rule

990 sub-codes



15 codes



4 themes

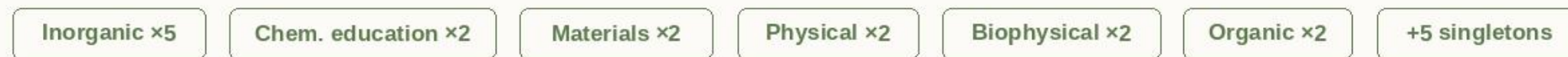
No new codes appeared after interview 7 — the result held from 10 participants to all 20.

THE SAMPLE · N = 20

Recruited for *maximum variation* across profile and subfield



SUBDISCIPLINES



Analytical generalization & transferability — not population estimates. Skews senior/male; see limitations.

THE FINDING

Faculty *agree* on the propositions.
They *divide* on where the boundaries fall.

“What differs is not which beliefs faculty hold, but where they draw the lines those beliefs require.”

The doctorate develops a self-directed scientist in service of the student's own goals.

“*When they start bringing ideas forward without necessarily being prompted.*

— P2 · on what independence looks like

WHERE FACULTY DIVIDE

Answering existing questions

Asking new questions

The realistic goal vs. the higher bar for the degree

Students learn by doing scaffolded research —
and learning shows in performance, not grades.

“*I do. We do. You do — spread out over X number of years.*”

— P5 · on how research is learned

WHERE FACULTY DIVIDE

Unsupported struggle helps

Struggle should be scaffolded

Same formative deprivation, opposite lessons — P13 vs. P7

Development is a shared responsibility in a two-way relationship — and who develops which skill varies.

“*I go from the leader to the follower, and they start as the follower and end up as the leader.*

— P6 · on how responsibility transfers

WHERE FACULTY DIVIDE

Faculty's to teach

Student's to build

The same skill — e.g. creativity — is one faculty's to teach (P5, P17), another's for the student (P9, P11)

Structure shapes advising — and the gaps it leaves are the program's to fix, not the student's.

“*Those professional skills often get kind of forgotten.*”

— P2 · on the program's gap

WHERE FACULTY DIVIDE

The evolved system works

Curricula & funding need change

How much of the structure deserves changing — P14 vs. P18

THE FULL OUTCOME SPACE

Within each shared theme, faculty differ along the dimensions below. Each axis names the two poles of variation; nothing is plotted on it.

Theme 1. The doctorate develops a self-directed scientist in service of the student's goals		
IC: Independent research directed primarily by you	← answering existing questions	asking new questions →
UN: Independent and useful preparation in science	← weight on independent science	weight on career preparation →
SG: Doctoral education as service to the student's goals	← career preparation as the defining purpose	career preparation as a minor benefit →
EE: The PhD as a foundation, not an endpoint	← independence largely reached by the time	the PhD as a foundation only →
Theme 2. Students learn by doing scaffolded research, shown in performance		
LE: Learning through experimental performance	← unsupported struggle is productive	struggle should be scaffolded →
LO: Learning shown in observable performance	← publication is a reliable indicator	publication is an unreliable indicator →
AV: Independence as active collaboration	← active checking	calibrated trust →
Theme 3. Development is a shared responsibility within a two-way relationship		
MB: Mentoring as a balanced relationship	← hands-on mentoring	hands-off mentoring →
FA: Faculty as primary agents of skill development	← faculty-guided development	self-directed development →
MP: Motivation as the most transferable responsibility	← motivation is innate	motivation is developed →
OE: Skill development as faculty responsibility only by engagement	← affective growth is a high priority	affective growth is less essential →
RS: Responsibility as shared and progressively transferred	← responsibility stays shared	responsibility fully transfers to the student →
SD: Skill development as the student's responsibility	← the capacity is built through initiative	the capacity is partly given →
Theme 4. Structure shapes advising and leaves gaps the program owns		
PS: Practice is shaped by structural conditions	← the evolved system mostly works	curricula and funding need real change →
GA: Graduated care and the program's responsibility	← professional and career gaps	teaching as a gap →

4 shared themes.

15 codes.

An axis of division in every one.

Every faculty member lands somewhere on each axis. The agreement is real — and so is the disagreement one level down.

Each axis names the two poles of variation; nothing is plotted on it.

A belief baseline for changing doctoral education

Content for the model

Gives the Teacher-Centered Systemic Reform model the doctoral-education content it lacked — the shared ground a reform can assume, and the fault lines it must engage.

Nests with prior work

The goals record (what the degree is for) and the challenge record (what goes wrong) now sit beside what faculty believe. Three answers, one enterprise — they nest, not compete.

Conviction vs. practice

13 of 20 describe stable convictions but implementation refined by experience. Supports aimed at implementation may fit faculty experience better than efforts aimed at core beliefs.

IMPLICATIONS

Two levels of the finding,
two lessons for programs.

Shared propositions

Common ground a reform argument can take as given — arguing from where faculty already stand.

Boundary disagreements

The harder, more useful target — where colleagues who sound aligned quietly diverge. Surface them, don't assume them.

Thank you.

Zoe Rahimi Pirkoohi & Jordan Harshman · University of Iowa

Supported by NSF CAREER #2142873 & #2602955 · Jordan Harshman, PI